STUDYLAB HUB · COMPUTER SCIENCE & ALGORITHMS · UNIT 1 (CHAPTERS 1–2)

# Unit 1: Big-O Complexity, Arrays & Sliding Window

Asymptotic Analysis O(1) to O($2^n$) · Amortized Dynamic Arrays · Prefix Sums · Two Pointers & Sliding Window

## Computer Science & Data Structures — Unit 1 (Chapters 1–2): Big-O Time/Space Complexity, Arrays, Two Pointers & Sliding Window

*Independent, original study guide for Computer Science & Data Structures (informed by OpenStax open textbooks, CC BY 4.0).*

### 1. Core Unit Roadmap

| Section | Core Principle & Definitions | High-Yield Exam Application |
|---|---|---|
| **Part 1 · Fundamental Concepts** | Asymptotic Analysis O(1) to O($2^n$) | Identifying core variables, boundary conditions, and conserved quantities |
| **Part 2 · Quantitative Models & Mechanisms** | Amortized Dynamic Arrays | Applying governing equations and predicting how system state changes under perturbation |
| **Part 3 · Advanced Problem Solving** | Two Pointers & Sliding Window | Avoiding common sign, unit, and conceptual traps on multiple-choice and free-response questions |

### 2. Essential Principles & Step-by-Step Analysis

1. **First-Principles Mechanism**:

- Every problem in **Unit 1 (Chapters 1–2): Big-O Time/Space Complexity, Arrays, Two Pointers & Sliding Window** reduces to identifying the underlying structure, forces, rates, or invariants before plugging in numbers.
- Always state the governing principle explicitly at the start of any free-response justification.

2. **High-Yield Comparison & Edge Cases**:

- Distinguish between equilibrium/steady-state behavior and transient/limiting behavior.
- Verify dimensional consistency and limiting cases (e.g., as distance or concentration approaches zero or infinity).

3. **Worked Exam Strategy**:

- Write out the known quantities with units, select the exact relation connecting them, solve symbolically first, and then substitute numerical values.